STATISTICAL SAMPLING

Terms and notation

A *sample* provides a set of data values of a random variable, drawn from all such possible values, the *parent population*. The parent population can be finite, such as all professional footballers, or infinite, such as the points where a dart can land on a dart board.

A representation of the items available to be sampled is called the *sampling frame*. This could, for example, be a list of the sheep in a flock, a map marked with a grid or an electoral register. In many situations no sampling frame exists nor is it possible to devise one, for example, for the cod in the North Atlantic. The proportion of the available items that are actually sampled is called the *sampling fraction*.

measure 10 out of 60 $\frac{10}{60}$

A parent population, often just called the *population*, is described in terms of its *parameters*, such as its mean, μ, and variance, σ^2. By convention Greek letters are used to denote these parameters.

μ mean population

A value derived from a sample is written in Roman letters: mean, $\bar{x}$, variance, s^2, etc. Such a number is the value of a *sample statistic* (or just *statistic*). When sample statistics are used to estimate the parent population parameters they are called *estimates*.

Thus if you take a random sample in which the mean is $\bar{x}$, you can use $\bar{x}$ to estimate the parent mean, μ. If in a particular sample $\bar{x} = 23.4$, then you can use 23.4 as an estimate of the population mean. The true value of μ will generally be somewhat different from your estimated value.

Upper case letters, X, Y, etc., are used to represent the random variables, and lower case letters, x, y, etc., to denote particular values of them.

Chosing an appropriate sample

- An appropriate sample should be as representative of the parent population as you can ~~make it~~ given the resources you have (time, money, etc). Every member of the population should have a non-zero probability of being selected. In many random sampling techniques, every member of the population has an equal probability of being selected.
- It should be free from bias. i.e unfairly representing part of the population.
- It should be sufficiently large for the results to have some meaning. (The smaller the sample, the higher the margin of error)

USING 'RANDOM NUMBERS' to select a sample.

Method

1. Give each member of the population a unique number
2. Using random number tables, choose random numbers of a suitable length If 0 - 9346 (4 digits) 0 - 50 (2 digits)
3. Enter members of the population matching these random numbers into your sample
4. If the same random number comes up more than once, ignore repetitions
5. Ignore random numbers that are larger than the size of the population.

Example :

A population has 463 members stop numbered 1 to 463. A sample of size 30 is required. Use the following random members to find the first three members of the sample.

3 digits - (463) 001 → 999

11164 36318 75061 37674 26320 75100 10431 20418
19928 91792

have to be consistent

Sample member 111, 363, 376

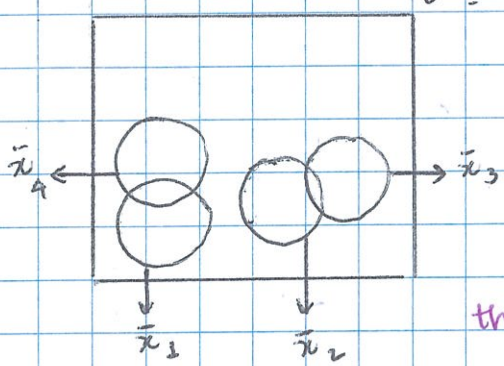
THE CENTRAL LIMIT THEOREM (CLT)

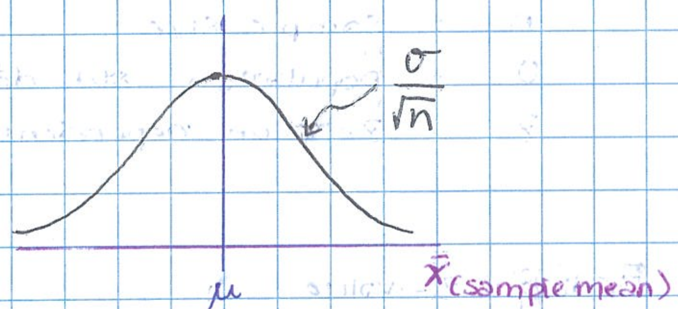
Given a parent population with mean μ and standard deviation σ, samples of size n will have a distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$

If n is large enough ($n \geq 30$), the distribution of sample means will be normal

(normal if $n \geq 30$)

Parent population $\mu =$
$\sigma^2 =$





the bigger the sample, less spread (σ^2)

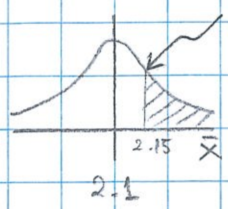
NOTE! If the parent population itself is normal, then the sample means will be normally distributed regardless of the sample size n.

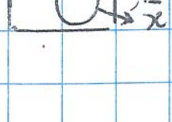
Eg. A population rats has a mean weight of 2.1 kg and a variance of 0.2 kg. The sample of 50 rats is weighted. What is the probability that the mean weight of the sample is bigger than 2.15 kg

the Q give σ^2 so, put $\sqrt{\ }$

$$\frac{\sigma}{\sqrt{n}} = \frac{\sqrt{0.2}}{\sqrt{50}} = 0.063245\ldots$$

Answer $n \geq 30$
$\mu = 2.1\text{kg}$
$\sigma^2 = 0.2\text{ kg}$





$$P(\bar{x} > 2.15) = P\left(Z > \frac{2.15 - 2.1}{\frac{\sqrt{0.2}}{\sqrt{50}}}\right)$$

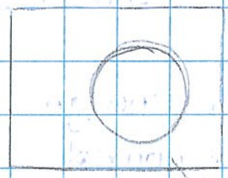
$$= P(Z > 0.79056\ldots)$$

$$= 0.2146 = 0.215$$

CONFIDENCE INTERVALS FOR A POPULATION MEAN

An α% confidence interval (CI) for the population mean μ is calculated from

$$\bar{x} \pm Z \frac{\sigma}{\sqrt{n}}$$



μ ?
σ ?

$n = \infty$

1.5 ⟷ 1.8

(CI : range)

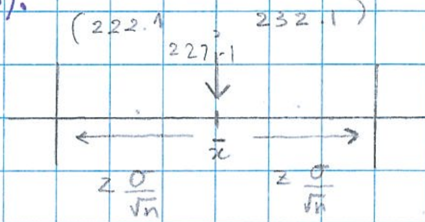
where

$\bar{x}$ = sample mean

n = sample size

σ = population std. dev.

Z = Z-value dependent on α%



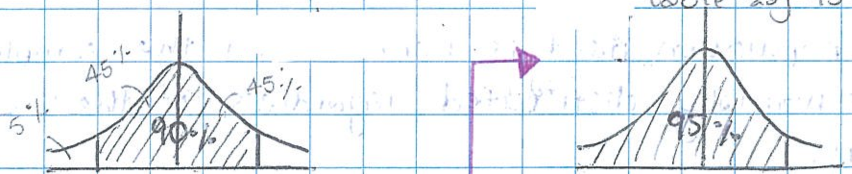
Finding Z - value

→ 95% of Z-value

e.g for a 90% CI.

$Z = 1.645$

table as 95%



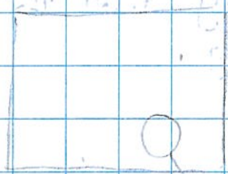
$Z = 1.645$ $\quad$ $Z = 1.645$

common values :

α%	Z-value
90%	1.645
95%	1.960
99%	2.576
97%	2.17

confidence interval

μ = ?
σ = ?



$104.3 \pm Z \frac{\sigma}{\sqrt{30}}$ → does not know this

$\bar{x} = 104.3$ → not same as μ

$n = 30$

so, find CI

$$S^2 = \frac{1}{n-1}\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right)$$

Example:

A sample of 30 students had a mean cross-country time of 21.6 mins
It is known that standard deviation of all student times is 2.1 mins
Find a 95% C.I for the mean time of all student cross country runners.

Answer: $\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$

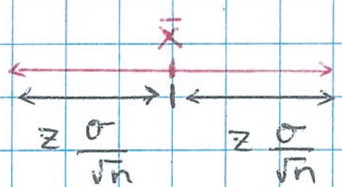
$= 21.6 \pm 1.960 \times \frac{2.1}{\sqrt{30}}$

$= [\, 20.8485\ldots,\ 22.35147\ldots \,]$

$= [\, 20.8,\ 22.4 \,]$ (3sf)

NOTE

The width of a CI



$$\text{width} = 2 \times z \frac{\sigma}{\sqrt{n}}$$

but, if Q says " within d "
then solve

$$d = z \frac{\sigma}{\sqrt{n}}$$

Relationship between width and n.

$$\text{Width} \propto \frac{1}{\sqrt{n}}$$

So, if n is increased by a factor of 4 width is halved $\left(\frac{1}{\sqrt{4}}\right)$

9 thirded $\left(\frac{1}{\sqrt{9}}\right)$

Unbiased Estimators

unbiased estimators for μ is.

$$\bar{x} = \frac{\Sigma x}{n}$$

unbiased estimators for σ^2 is

$$S^2 = \frac{1}{n-1}\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right)$$

$\bar{x}$, S^2 - for sample. μ, σ^2 population parameters.

E.g. Data 1, 2, 3, 4

Find unbiased estimators for μ and σ^2

$$\bar{x} = \frac{1+2+3+4}{4} = 2.5$$

x	x^2
1	1
2	4
3	9
4	16
Σx	Σx^2
10	30

$$S^2 = \frac{1}{3}\left(30 - \frac{(10)^2}{4}\right)$$

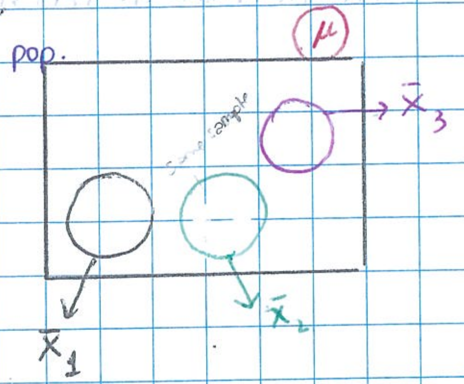
$$S^2 = \frac{5}{3} = 1.666\ldots$$

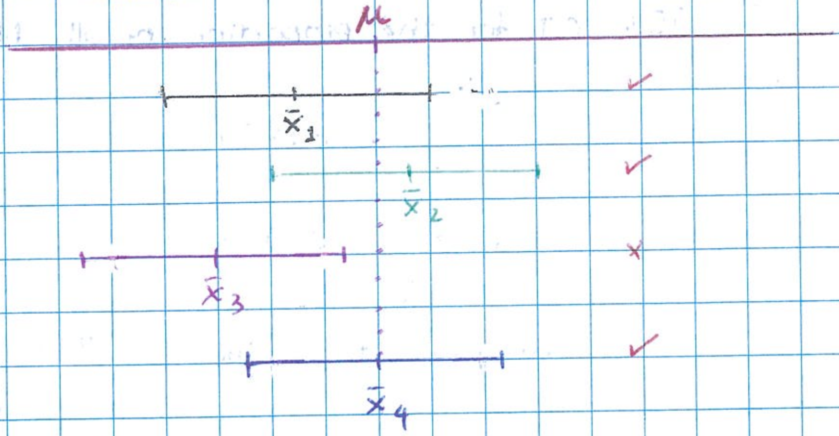
$$S = \sqrt{\frac{5}{3}} = 1.29099\ldots$$

NOTE: In the CI formula, if σ is not known, use S

$$\bar{x} \pm z \frac{\cancel{\sigma}}{\sqrt{n}} \quad S$$

How often does a CI "catch" the population mean

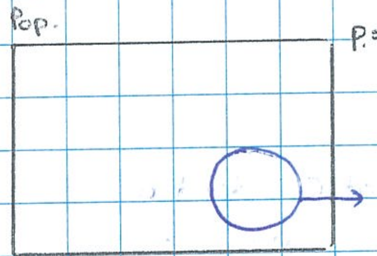




In theory, for α%, confidence intervals α%. will catch the true value μ

90% catch it, 10% doesn't catch

C.I FOR POPULATION PROPORTION



P = proportion of human that are left handed

P_s = Sample proportion

Sample proportion
yes / NO
left hand

An α% CI for the population proportion

$$P_s \pm z\sqrt{\frac{P_s(1-P_s)}{n}}$$

E.g. At strathallan 114 of 700 students are left-handed. Find a 95% C.I for the proportion of all NZ students that are left handed

Ans:

$$P_s = \frac{114}{700} = 0.162857$$

CI $\quad P_s \pm Z\sqrt{\frac{P_s(1-P_s)}{n}}$

$$= 0.162857 \pm 1.960\sqrt{\frac{0.162857(1-0.162857)}{700}}$$

$$= [0.1355036..., 0.1902103...] = [0.136, 0.190] \text{ 3sf}$$

Sample of n, 61 of them read the reporter newspaper ... 9709/P7/LQ3 June 2005

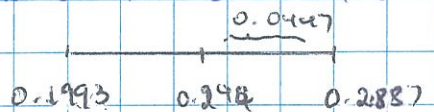
E.g. mid-point of C.I $\quad 0.1993 < p < 0.2887$

$$= \frac{0.1993 + 0.2887}{2} = 0.244$$

find n $\quad \frac{61}{n} = 0.244$

$$n = 250$$

Find confidence level of C.I or α% of C.I find α

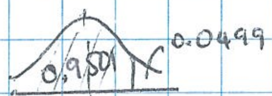


α% of CI

$$0.0447 = Z \times \sqrt{\frac{P_s(1-P_s)}{n}}$$

$$0.0447 = Z \times \sqrt{\frac{0.244(0.756)}{250}}$$

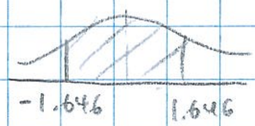
$$Z = 1.646$$
$$= \phi(0.9501)$$



$$1 - 2(0.0499)$$
$$= 0.9002$$

so, $\alpha = 90\%$

or



$$P(-1.646 < z < 1.646) = (0.9501) - (1 - 0.9501)$$